

CHAPTER 1

INTRODUCTION

In the 21st century, computer technology has become the bedrock of modern society, seamlessly integrated into nearly every facet of our daily lives. From communication and education to business and entertainment, computational systems have revolutionized how we work, learn, and interact. This digital transformation has established a new paradigm where efficiency, accessibility, and connectivity are paramount, fundamentally reshaping human experience and expectation. The field of multimedia and interactive software stands as a prime example of this evolution, pushing the boundaries of creativity, storytelling, and user engagement.

Within this digital landscape, the domain of game development exemplifies the powerful convergence of art and technology. Computerization in this field has enabled the creation of rich, immersive worlds and complex systems that challenge and entertain millions. Specifically, the survival-strategy genre leverages computational power to simulate intricate ecosystems of resource management, strategic planning, and dynamic AI behavior, offering players a deeply engaging test of foresight and adaptability. The design of these systems is guided by established principles of human-computer interaction (HCI) and cognitive load theory, which emphasizes that effective software must not only be powerful but also intelligible to the user. Theories of player onboarding, such as those discussed by Benvenuti et al. (2023) and Chen (2021), argue that learning is most effective when it is contextual and interactive, moving beyond passive reading to active, guided doing. A well-designed game, therefore, acts as a

sophisticated system that teaches its own mechanics through experience, balancing depth with clarity.

However, a significant problem persists within this advanced ecosystem. Many survival-strategy games, in their pursuit of depth and complexity, create a steep learning curve that alienates a substantial portion of their potential audience. New players are often confronted with an overwhelming array of mechanics, managed through complex control schemes, and explained through inadequate, text-heavy tutorials. This conventional approach fails to adhere to the very principles of effective learning it seeks to employ, leading to cognitive overload and player frustration (Luna, 2025). The consequence is a critical accessibility barrier: the very players who might enjoy the strategic core of the game are discouraged from engaging with it at all. This research directly addresses this issue by developing "Bastion of Genesis," a game designed to solve the onboarding problem through streamlined controls and an enhanced, interactive tutorial system, thereby making the strategic depth of the genre accessible to both new and experienced players alike.

1.1 Background of the study

The survival-strategy genre occupies a unique and cherished space in the digital entertainment landscape, captivating players with its blend of immediate survival pressures and long-term strategic planning. However, for years, a significant challenge has shadowed this genre: a notoriously steep learning curve. New players, eager to experience the depth these games offer, often find themselves confronted by a wall of complex mechanics, convoluted control schemes, and tutorials that fail to adequately prepare them for the

challenges ahead. This onboarding crisis is not just anecdotal; it is a well-documented issue in game design research. Studies consistently show that conventional tutorial methods, such as walls of text, often lead to cognitive overload, while detached, simplistic tutorials fail to engage players, resulting in higher drop-out rates (Benvenuti et al., 2023; Chen, 2021; Luna, 2025).

The proponents and developers of this project have observed this problem firsthand. As passionate players of strategy games themselves, the team recognizes that their familiarity with genre conventions is not shared by everyone. Friends and potential players often become discouraged and abandon fantastic games not for a lack of interest, but because the initial barrier to entry is simply too high.

It is from this intersection of personal observation and established research that our project, “Bastion of Genesis,” was born. The proponents selected this specific game concept precisely because its core innovations naturally present a complex onboarding challenge, making it an ideal candidate to test our solutions. The game introduces a dynamic Pollution-Hostility System, where industrial expansion directly aggravates the native fantasy races, and a Strategic Triad that forces constant trade-offs between expansion, economy, and defense. These systems are the source of the game's strategic depth, but they are also precisely the kind of mechanics that can overwhelm a new player if not introduced carefully.

Therefore, the justification for this study rests on two interconnected goals. First, it emerges from a genuine desire to contribute to the evolution of

the survival-strategy genre, helping it grow by becoming more welcoming and inclusive. Second, it is driven by a concern for improving game development practices. Through this project, the researchers are conducting a practical investigation into how to balance accessibility with depth. "Bastion of Genesis" serves as a case study to demonstrate that specifically thoughtful design, streamlined controls paired with an interactive, context-sensitive tutorial system can lower the barrier to entry without compromising the rich strategic experience that defines the genre. This research aims to transform a prevalent industry problem into an opportunity for innovation, creating a blueprint for future developers, and a more engaging experience for players everywhere.

1.2 Objectives of the study

1.2.1 General Objective

To create a more welcoming survival-strategy game that balances accessibility with depth. This will be achieved by designing intuitive controls and a seamlessly integrated tutorial system, guided by established principles of player onboarding, to lower the initial barrier for newcomers without compromising the complex strategic decisions.

1.2.2 Specific Objectives

- To design and implement streamlined gameplay mechanics that let players manage survival resources efficiently, reducing unnecessary complexity while maintaining strategic depth.

- To create an improved, easy-to-follow tutorial system that gradually introduces survival-strategy concepts, enabling players to learn and adapt without sacrificing challenge.
- To assess the effectiveness of the user-friendly controls and enhanced tutorial design in making the game more accessible and engaging for both new and experienced players.

1.3 Significance of the study

1.3.1 Contribution in the field

This project is significant as it addresses one of the most common barriers in strategy-based games, the steep learning curve caused by complex controls and insufficient tutorials. By creating a survival-strategy game with simplified mechanics and an improved tutorial system, the study contributes to making the genre more accessible to a wider audience, including beginners who are often discouraged from playing such games. This project adds to the field of game development by exploring design practices that balance accessibility with depth, serving as a reference for future developers who aim to create engaging yet user-friendly strategy games. In doing so, it promotes innovation in gameplay design, user experience, and player engagement within the broader domain of game development research.

1.3.2 Beneficiaries of the study

The outcome of the study will be beneficial to the following:

- New and Beginner Players: This group is the primary beneficiary.

The research directly addresses the steep learning curve and

insufficient tutorials that often discourage them. By creating a more accessible game with an "improved tutorial system, the project aims to lower the barrier to entry and make the survival-strategy genre more welcoming to them.

- **Game Developers:** The project serves as a practical reference for future developers. By exploring design practices that balance accessibility with depth, it provides valuable insights and a framework that developers can use to create their own "engaging yet user-friendly strategy games," thereby saving time and resources on research and development.
- **The Game Development Research Field:** The study adds to the field of game development. Its contributions to knowledge in gameplay design, user experience, and player engagement benefit academics and researchers by providing a case study and advancing the understanding of how to create more effective and inclusive game designs.
- **The Survival-Strategy Genre:** The broader genre benefits from this research. By demonstrating a successful model for combining simplified mechanics with strategic depth, the project promotes innovation and helps evolve the genre to be "more accessible to a wider audience, ensuring its longevity and relevance.

1.4 Scope and Limitations

- The project will deliver a 2D top-down survival strategy game with 50 levels divided into 5 progressively harder chapters.

- The game will feature LAN Multiplayer (2-Player MAX) for cooperative play via unified, shared base state and asynchronous input control for each player via independent cursors. This configuration establishes a purely collaborative environment within a shared operational domain.
- The core gameplay loop includes facility construction, resource gathering, pollution management, research progression, and defense against hostile factions.
- The game will include five major facility types (e.g., mining, energy, research, defense, production).
- Enemy factions will consist of four races: Humans, Elves, Dwarves, and Demons, each with distinct attack styles and escalation patterns.
- The pollution-hostility system will be implemented to control difficulty and narrative tension.
- Reward-Driven Progression is structured as a staircase of increasing complexity; each mission cleared serves as the key to activating the next tier of construction and resource acquisition.
- Enemy Composition is structured sequentially across chapters, serving as a controlled increase in complexity and challenge:
- Chapter 1: Human Enemies Only – Introduced as the most basic enemy dealing only physical damage.

- Chapter 2: Human and Dwarves are introduced to make players change strategies to two distinct enemy profiles; Dwarves can go underneath and inside walls.
- Chapter 3: Dwarf and Demon – Demons are introduced as the first airborne magic-type enemies that deals with destructive power to the player's camp.
- Chapter 4: Demon and Elf – Elves are the last enemy to be introduced as magic-type enemies who can attack and heal while lacking significant health.
- Chapter 5: All Enemies are present as the last wave of enemies to challenge the player demanding mastery of all previously learned strategies.

1.5 Conceptual Framework

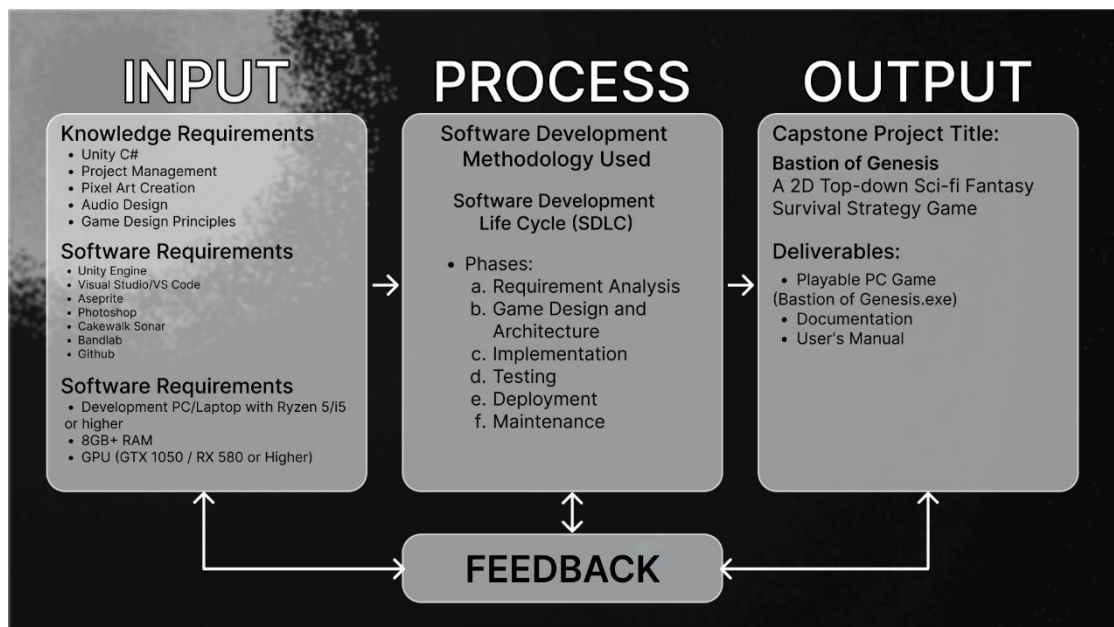


Figure 1. IPO Diagram

CHAPTER 2

REVIEW OF THE RELATED LITERATURE

This chapter explores the existing knowledge that forms the foundation for our game development.

2.1 Related Literature

2.1.1. Foreign Literature

The exploration of international design literature provides the fundamental principles guiding our approach to game design. The Interaction Design Foundation's perspective on Human-Computer Interaction establishes that technology should serve human needs rather than force users to adapt to complex systems (Interaction Design Foundation, n.d.). This philosophy became central to our design process, influencing decisions from interface layout to tutorial structure, ensuring that player experience remained the priority in every development choice.

Building on this foundation, Jakob Nielsen's Usability Heuristics offer practical guidance for creating interfaces that feel intuitive rather than foreign (Nielsen, 1994). The principle of "matching the system to the real world" guided our use of clear, accessible language instead of specialized gaming jargon, while "recognition rather than recall" inspired our implementation of visual tooltips and persistent information displays. These design choices allow players to focus on strategic thinking rather than command memorization, creating a more fluid and engaging gameplay experience.

The cognitive aspects of game design find strong support in the Education Endowment Foundation's work on Cognitive Load Theory, which

provides scientific validation for the intuitive understanding that information overload impedes learning (Education Endowment Foundation, 2021). This theoretical framework directly shaped our tutorial system, where game mechanics are introduced progressively, enabling players to achieve mastery of each concept before encountering additional complexity. This structured approach to information delivery ensures that new players can build confidence while gradually developing the skills needed for advanced strategic play.

The human-centered design philosophy championed by IDEO emphasizes the importance of developing genuine empathy with end-users (IDEO, 2015). By creating detailed player personas representing both strategy veterans and complete newcomers, the proponents maintained consistent focus on building accessible pathways for players encountering strategy game concepts for the first time. This practice consistently prevented the common pitfall of designing for our own preferences as experienced gamers, instead keeping genuine player needs at the forefront of development decisions.

Complementing these approaches, the Game Accessibility Guidelines demonstrate that designing for accessibility ultimately creates superior experiences for all players (Game Accessibility Guidelines, 2022). This perspective validates our core belief that clear tutorials and streamlined controls represent not a "dumbing down" of game complexity, but rather an opening up of strategic depth to wider audiences. By removing unnecessary barriers to entry, more players can be invited to appreciate the rich tactical possibilities that survival-strategy games offer.

2.1.2. Local Literature

The examination of Philippine digital landscape and policy context reveals both the opportunity and necessity for accessible game design in the local market. Statistical data from Statista identifies the Philippines as one of Southeast Asia's fastest-growing gaming markets, representing millions of potential players who deserve gaming experiences that respect their time and intelligence while welcoming them into new genres (Statista, 2023). These figures represent not just business opportunities but real people - friends, family members, and students - who stand to benefit from more accessible game design.

National policy frameworks further reinforce the importance of this development direction. The Department of Information and Communications Technology's Digital Transformation Strategy outlines a vision for fostering vibrant ICT-based content industries, positioning projects like Bastion of Genesis as contributions to national digital development goals (Department of Information and Communications Technology, 2022). Similarly, the Intellectual Property Code provides crucial protection for creative works, ensuring that original game mechanics and artistic vision can be recognized as legitimate contributions to both the gaming industry and cultural landscape (Republic of the Philippines, 1997).

Industry analysis from publications like Esquire Philippines paints a picture of a local game development community finding its distinctive voice while navigating the challenge of creating products with both local relevance and international appeal (Esquire Philippines, 2023). This context highlights the importance of developing design approaches that can resonate across cultural boundaries while addressing universal player needs. The broader educational

context, as reflected in Commission on Higher Education policies promoting digital literacy, suggests that well-designed games can serve dual purposes as both entertainment and vehicles for developing valuable cognitive skills (Commission on Higher Education, 2019).

2.2 Related Studies/Projects

2.2.1. Foreign Related Projects

Several international games provide essential blueprints for the systems and design philosophy central to our project. *Eco* (Strange Loop Games, 2018) stands as a foundational case study in systemic consequence, where player-driven industry and pollution dynamically affect a shared world. This design directly validates the core mechanic of our Pollution-Hostility System, demonstrating that strategic choices can carry tangible narrative and mechanical weight.



Figure 2. Environmental systems in *Eco*



Figure 3. Environmental systems in Eco



Figure 4. Environmental systems in Eco



Figure 5. Environmental systems in Eco

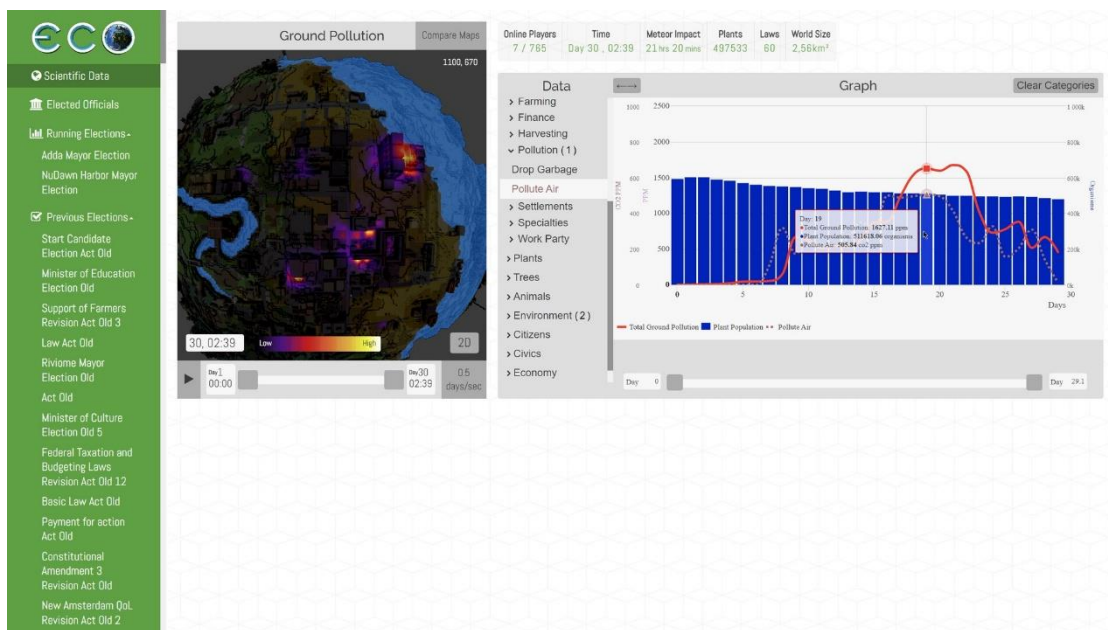


Figure 6. Environmental systems in Eco

For the intense, wave-based strategic tension we aim to create, *They Are Billions* (Numantian Games, 2019) serves as a critical reference. Its gameplay perfectly encapsulates the Expansion vs. Defense axis of our Strategic Triad, offering a masterclass in pacing and threat management.



Figure 7. Base defense mechanics in *They Are Billions*



Figure 8. Base defense mechanics in *They Are Billions*



Figure 9. Base defense mechanics in They Are Billions



Figure 10. Base defense mechanics in They Are Billions

In pursuit of complex, emergent storytelling, Rimworld (Ludeon Studios, 2018) is an unparalleled benchmark. Its AI-driven narrative system, where character psychology and random events weave unique stories, inspires our approach to creating dynamic interactions within the Tech vs. Magic conflict.



Figure 11. Emergent Story-Telling in RimWorld



Figure 12. Emergent Story-Telling in RimWorld



Figure 13. Emergent Story-Telling in RimWorld



Figure 14. Emergent storytelling interface in Rimworld

Forever Skies (Far From Home, 2025) demonstrates atmospheric world-building that integrates survival mechanics into exploration, showing how environmental storytelling can create a compelling player journey.



Figure 15. Atmospheric exploration in Forever Skies



Figure 16. Atmospheric exploration in Forever Skies

Finally, Survival Kingdoms (BatByte, 2016) presents a streamlined approach to kingdom management, directly aligning with our project's goal of lowering the genre's barrier to entry.



Figure 17. Streamlined interface design in Survival Kingdoms



Figure 18. Streamlined interface design in Survival Kingdoms

2.2.2 Local Related Projects

The Philippine game development landscape offers a range of projects that provide both practical and inspirational guidance, demonstrating the feasibility and creative potential of local production. *Anito: Defend a Land Enraged* (Anino Entertainment, 2003) holds historical significance as one of the first commercially released Filipino-made PC games. Its deep integration of pre-colonial mythology proves the cultural resonance and commercial viability of original Filipino intellectual property, encouraging our pursuit of a unique game world.

More recent titles highlight the evolving technical and creative standards of the industry. *Garrison: Archangel* (Indigo Entertainment, 2020), a mecha-based real-time strategy game, showcases the capacity of local studios to execute polished, original titles in a complex genre. Similarly, *Conquered* (Lordd Lazaro, 2017) and *Nightfall: Escape* (Zeenoh Inc., 2016) represent Filipino-led ventures into strategy and action-survival, respectively, proving that local developers can successfully navigate genre conventions and produce engaging experiences. Finally, the student-developed project *Insureksyon* (ADNU DACA Games, 2025) provides the most direct parallel to our own context as an academic capstone. It demonstrates how a university-level project can tackle serious themes while developing core gameplay, serving as a relevant model for scoping and executing a meaningful game within an academic framework.

2.3 Synthesis

A detailed analysis of related literature and existing game systems reveals two powerful, yet often separate, trajectories in game creation. On one trajectory, international genre-defining games set the standard for depth and engagement. Titles like *Eco* by Strange Loop Games (2018) and *Rimworld* by

Ludeon Studios (2018) demonstrate the compelling potential of complex, interlocking systems that foster strategic thinking and emergent storytelling. Their design success is implicitly supported by established human-computer interaction principles. These principles, such as the usability heuristics that prioritize clear and intuitive interfaces, form the theoretical foundation for creating digital experiences that are both powerful and navigable. These international benchmarks effectively illustrate the destination, which is rich strategic gameplay, and the proven design principles that serve as the map for getting there.

The second trajectory is the vibrant growth of the Philippine game development industry. This is evidenced by pioneering works like *Anito: Defend a Land Enraged* from Anino Entertainment (2003) and contemporary projects such as *Garrison: Archangel* by Indigo Entertainment (2020). This local ecosystem showcases a clear evolution of technical skill, creative storytelling, and the confidence to build original worlds. Filipino developers are capably mastering complex genres and proving that high-quality, culturally-informed games can originate from the local scene.

However, placing these two trajectories side by side reveals a significant and actionable gap. The international titles that achieve the greatest strategic complexity often present a steep initial learning curve. This difficulty can deter new players, creating a barrier that contradicts the very principles of accessible and user-centered design. Meanwhile, the local industry's exploration of similarly deep, system-driven strategy games is still developing. This presents a clear opportunity: to create a project that consciously bridges this divide. The gap, therefore, is the absence of a demonstrated model that seamlessly integrates

sophisticated, emergent strategy with a foundational and deliberate commitment to accessible onboarding. This model would be specifically conceived and executed within the context of Philippine game development.

This identified gap is the core challenge that Bastion of Genesis is designed to address. The project is distinct in that it functions as a practical design intervention rather than simply another entry in the survival-strategy genre. It constitutes a hands-on investigation into whether the systemic depth found in games like Rimworld can be merged with the accessibility and guided learning emphasized by human-computer interaction research. In this approach, complex mechanics like the Pollution-Hostility System are not hurdles to be lowered but intricate systems to be expertly introduced. The tutorial and user interface are treated as primary design components, with the explicit goal of making profound strategic concepts understandable and engaging from a player's first moments.

The relevance of this focused synthesis to the project is fundamental. Bastion of Genesis is the practical test of our research hypothesis. It serves as a live case study examining whether deep strategic gameplay and genuine accessibility can work in harmony. Through meticulous documentation of the process, including the application of cognitive load theory to tutorial structure and the implementation of usability standards in the interface, this research aims to contribute more than a final product. The intended outcome is a documented methodology and a set of applicable techniques for other developers. Ultimately, this work aspires to support the local industry by demonstrating that Filipino teams can not only create games with international-caliber strategic

depth but can also pioneer more inclusive ways to welcome players into complex worlds, thereby offering a valuable contribution to global game design practices. Thereby enriching the global gaming landscape with more inclusive design.

2.4 Definition of Terms

Accessibility (in Game Design): The design of game systems, interfaces, and mechanics to be usable by the widest possible range of players, including those with disabilities or those who are new to a genre, by removing unnecessary barriers to entry and play.

Bastion of Genesis: The title of the survival-strategy game developed in this study, thematically representing a stronghold created at the origin of a new world.

Cognitive Load: The total amount of mental effort being used in the working memory, where excessive load can impede the acquisition of new skills and information.

Human-Computer Interaction (HCI): A field of study focused on the design and use of computer technology, centered on the interfaces between people and computers, with an emphasis on usability and user experience.

Onboarding (Player Onboarding): The process of integrating a new player into a game, during which they learn the core mechanics, controls, and goals.

Pollution-Hostility System: A core game mechanic where a player's industrial activities generate pollution, which directly increases the aggression and attack frequency of indigenous factions within the game world.

Survival-Strategy Genre: A video game genre that combines the resource management and long-term planning of strategy games with the constant pressure of maintaining vital resources against a hostile environment.

Strategic Triad: A game design model creating a constant, interconnected trade-off between three core pillars: Expansion, Economy, and Defense, where advancement in one area creates challenges in the others.

User Interface (UI): The space where interactions between humans and machines occur; in games, it includes all on-screen elements that provide information to the player and allow them to interact with the game world.

Usability: A quality attribute that assesses how easy user interfaces are to use, based on principles of learnability, efficiency, memorability, and error prevention.

CHAPTER 3

TECHNICAL BACKGROUND AND METHODOLOGY

This chapter details the technical architecture of Bastion of Genesis, documenting the complete development ecosystem from conception to deployment. It outlines our research design and software development methodology, explaining our structured approach to building the game.

3.1 Technical Background

3.1.1 Technicality of the project

The development of Bastion of Genesis is supported by a purposefully selected suite of technologies, chosen by the proponents to effectively bring the project from concept to a functional game.

Unity Engine provides the foundational framework that houses the entire project, enabling efficient 2D development and cross-platform deployment. For programming, **Visual Studio Code** offers comprehensive debugging tools for complex systems, it provides an integrated development environments for C# scripting and lightweight alternative for quick edits and version control operations. Our creative pipeline is powered by specialized tools: **Aseprite** delivers precise pixel art creation for our game's visual identity, while Photoshop handles sophisticated UI design and promotional assets. The auditory experience is crafted through **Cakewalk Sonar** for professional audio production and **Bandlab** for collaborative sound design. Finally, **GitHub** serves as the project's backbone, ensuring version control and seamless team collaboration throughout our development journey. This

harmonious integration of technologies creates a robust foundation that supports both our creative vision and technical requirements from conception to completion. **Details of the technologies used**

In developing the project, the following technology tools will be used:

Unity Engine - serves as our primary game development framework, providing robust 2D development tools, physics systems, and cross-platform deployment capabilities that form the foundation of our project.

Visual Studio Code - provides a lightweight alternative for quick script editing and version control operations, complementing our primary IDE with its flexibility and extensibility.

Aseprite - specializes in pixel art creation and animation, offering precise tools for sprite development, palette management, and frame-by-frame animation that define our game's visual style.

Photoshop - handles complex asset creation including user interface design, promotional materials, and texture composition, providing advanced image editing capabilities for our visual elements.

Cakewalk Sonar - serves as our professional digital audio workstation for composing original music, designing sound effects, and implementing adaptive audio systems that respond to gameplay dynamics.

Bandlab - provides a collaborative audio platform for team-based sound design, voice-over editing, and audio asset management through its cloud-based workflow environment.

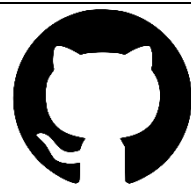
GitHub - forms our version control backbone, enabling team collaboration through branching strategies, pull request workflows, and project management tools that ensure code quality and development organization.

Table 1. Hardware Requirements for Development

Hardware Image	Name/Description/Model	Purpose in the Project	Price (original price and/or discounted price)	Availability in the Phil.
	Processor: AMD Ryzen 5 3400G with Radeon Vega Graphics 3.7GHz	It offers multi-threading capability, which allows the system to efficiently run game engines like Unity and support the complex processing tasks required for rendering, simulation, and real-time adjustments.	₱4,799	Shopee Lazada Pc Gilmore Computer Center
	RX 580 (8GB)	The RX 580 is a mid-range GPU with 8GB of memory, crucial for rendering high-quality graphics and managing the visual complexity of Bastion of Genesis. It enables the development and testing of game environments, character models, and smooth animations, even with moderately high graphical settings.	₱4,849	Shopee Lazada Pc Gilmore Computer Center
	16 GB DDR4 RAM 3200Mhz	This amount of memory allows the system to efficiently manage multiple applications and large assets (textures, animations, etc.), which are common	₱3,152	Shopee Lazada Pc Gilmore Computer Center

		in game development. It also supports a stable workflow, reducing crashes or slowdowns when working with large files and complex scenes.		
	1TB SATA SSD	provides ample storage and rapid data access, crucial for game development.	₱1,610	Shopee Lazada Pc Gilmore Computer Center
	24-Inch Monitor	A 24-inch monitor provides an adequate workspace for detailed viewing of game scenes, assets, and UI elements. The screen size allows for a comfortable multi-window setup, useful for running Unity and related software side by side.	₱1,589	Shopee Lazada Pc Gilmore Computer Center
	Mouse and Keyboard	Both devices contribute to a smooth workflow and allow for effective interaction with the software, which is essential for productivity and detailed adjustments in development.	₱254	Shopee Lazada Pc Gilmore Computer Center
			TOTAL COST	₱16,253

Table 2. Software Requirements for Development

	Software/App Name	Purpose in the Project	Software/App Logo
<i>Toolchain:</i>			
	Asperite	specializes in pixel art creation and animation	
	Photoshop	handles complex asset creation including user interface design, promotional materials, and texture composition	
	Visual Studio Code	provides a lightweight alternative for quick script editing and version control operations	
	Cakewalk Sonar	serves as our professional digital audio workstation for composing original music, designing sound effects.	
	Bandlab	Provides a collaborative audio platform for team-based sound design, voice-over editing, and audio asset management.	
<i>Engine:</i>			
	Unity Engine	serves as our primary game development framework, providing robust 2D development tools, physics systems, and cross-platform deployment capabilities that form the foundation of our project.	
<i>Other Libraries (if any):</i>			
	Github	forms our version control backbone, enabling team collaboration through branching strategies, pull request workflows, and project management tools that ensure code quality and	

		development organization.	
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3.1.2 Requirements Specifications

- **Gantt Chart**

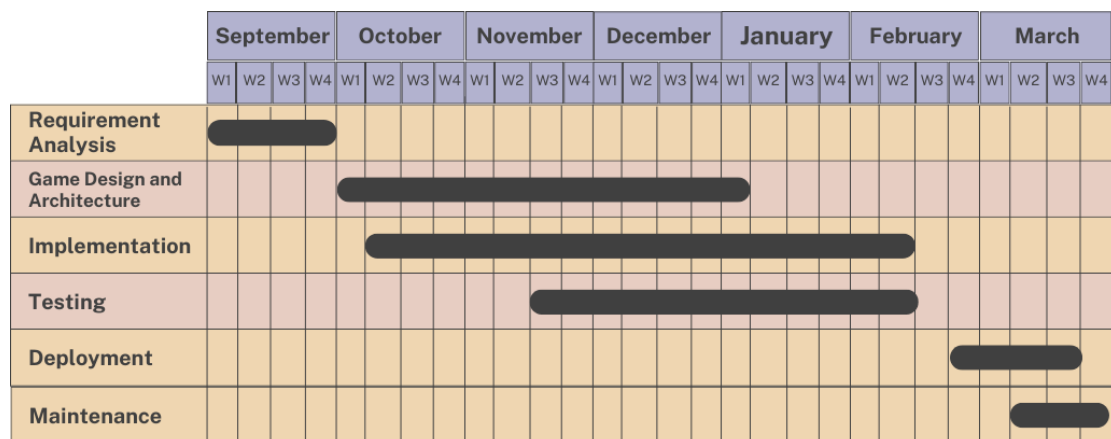


Figure 19. Gantt Chart

The Gantt chart for Bastion of Genesis outlines a structured project timeline following an iterative and overlapping workflow.

Requirement Analysis: Conducted over four weeks in September, this foundational phase is dedicated to defining the core mechanics, establishing the target scope, and finalizing all system requirements for the game. This ensures a complete and validated understanding of the project's objectives before development commences.

Game Design and Architecture: Spanning from the first week of October through the fourth week of December, this lengthy phase focuses on conceptualizing the complete game experience, developing the system architecture, and drafting the technical specifications. It concludes with the finalization of the Strategic Triad model and the initial design of the Destructive Terraforming system.

Implementation: Running concurrently from the second week of October to the second week of February, this is the core development period. It involves the coding of all game mechanics, integration of the Fish-Net LAN multiplayer system, and the

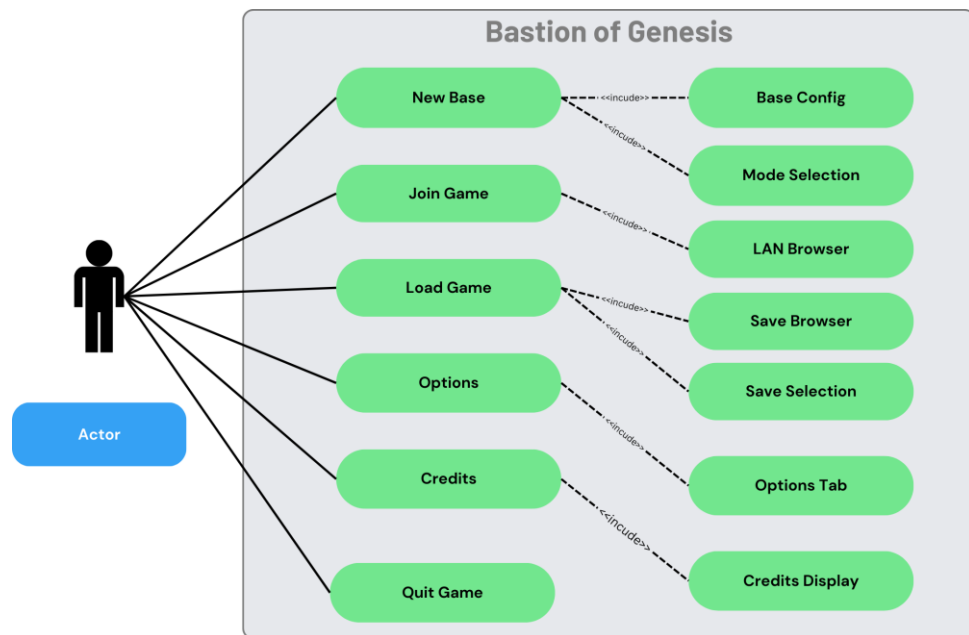
construction of the base building and combat systems. The long duration ensures ample time for coding complex interactions.

Testing: Overlapping heavily with implementation, starting in the third week of November and lasting until the third week of February, this phase involves rigorous quality assurance, bug fixing, and balancing of game mechanics. Starting early allows for immediate feedback loops and includes dedicated testing of the LAN cooperative mode to ensure stability.

Deployment: From the fourth week of February to the third week of March, this final preparatory phase involves packaging the final build, comprehensive documentation, and configuring the environment for the final thesis presentation or release.

Maintenance: Taking place from the second week of March until the fourth week of March, the concluding phase is reserved for post-deployment stabilization, addressing any critical issues discovered during the final review, and ensuring the long-term viability of the project framework.

Each phase is carefully scheduled to allow sufficient time for thorough completion.



- **Use Case Diagram**

Figure 20. Use Case Diagram

The Use-Case Diagram for **Bastion of Genesis** outlines the critical interactions between the primary actor (the Player) and the game's core functionality. Unlike standard single-player games, this diagram emphasizes the dual focus on solo progression and dedicated cooperative LAN play

Story Mode: The Story Mode serves as the single-player campaign. This use case encapsulates the entire solo experience, where the Player assumes the role of Kyra-Dominia. Key actions include establishing the base, executing the Destructive Terraforming protocol, and managing the resource-hostility loop autonomously. This path provides the foundation for learning the complex systems before cooperative play.

Create Game Server (LAN): This use case is the entry point for cooperative play. It allows a Player to establish a local, dedicated game session using the Fish-Net architecture. The initiating Player acts as the host, managing the game state and synchronization for other connected players. This setup is crucial for enabling the Shared-State Defense model, where the Strategic Triad balance is managed by a team.

Join Game Server (LAN): This use case enables other Players on the Local Area Network to connect to an existing server created via the **Create Game Server** use case. Upon successful connection, the joining Player is granted access to the shared base infrastructure and contributes to the economy, defense, and expansion efforts. This use case directly supports the thesis's focus on cooperative defense against the escalating Hostility mechanic.

- **Visual Table of Contents (VTOC) Diagram**

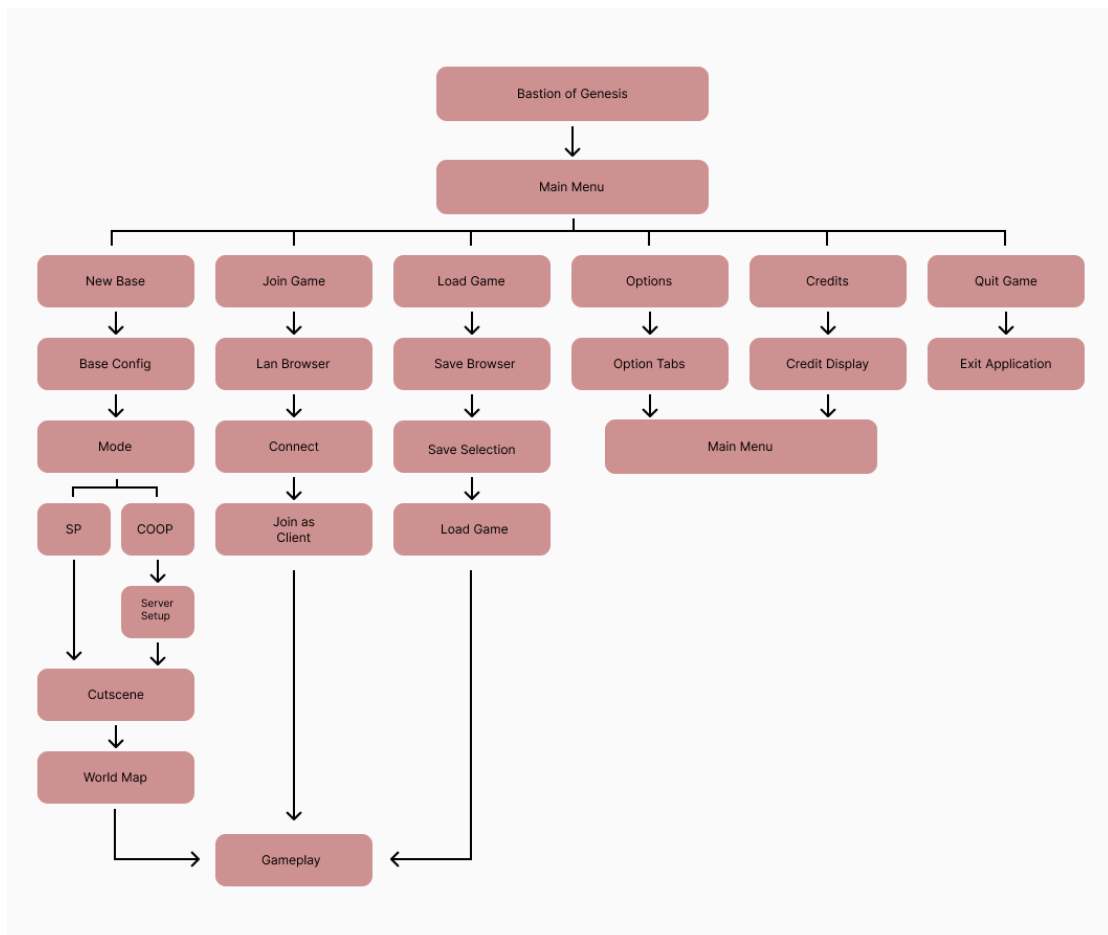


Figure 21. VTOC Diagram

The Visual Table of Contents (VTOC) Diagram for Bastion of Genesis offers an organized view of the game's main menu structure, laying out the

primary interactive elements in a visually accessible way. This structure allows players to see their options clearly and navigate smoothly.

3.2 Methodology

3.2.1 Research Design

This study is built upon a Developmental Research Design. This methodology is perfectly suited for projects, like ours, that aim to create a solution to a defined problem.

this research design allows us to function as both creators and investigators. It elevates Bastion of Genesis from a standard capstone project to a meaningful case study.

The developmental process ensures iterative building, progressing through clear phases of planning, construction, testing, and refinement. This structure allows for the gathering of concrete evidence at each stage, using feedback from both technical experts and target players to validate whether the designs successfully lower the barrier to entry. The project involves not only developing a game but also documenting a methodical process for balancing complexity with accessibility. This contributes practical, evidence-based insights to the field of game design that extend well beyond the prototype itself.

3.2.2 Software Development

The project will follow a structured Systems Development Life Cycle, specifically implementing the Spiral model. This approach was selected for its emphasis on iterative development and systematic risk management, which align perfectly with our project's complex, interconnected systems and research-driven nature.

The Spiral model, originally conceived by Barry Boehm, guides development through a series of iterative cycles (spirals), each consisting of four distinct phases: determining objectives, identifying and resolving risks, development and testing, and planning the next iteration. This structure is ideal for the research for three main reasons.

First, the project introduces complex, interconnected systems such as the Pollution-Hostility System and the Strategic Triad. These innovative mechanics carry inherent risks of imbalance, technical failure, or player confusion. The Spiral model's explicit focus on risk analysis at the beginning of each cycle allows us to identify potential issues early—before they become deeply embedded in the codebase—and develop mitigation strategies. This systematic approach to risk ensures that these complex systems work harmoniously and achieve their intended design goals.

Second, the core objective of balancing accessibility with strategic depth requires continuous refinement based on player feedback. The iterative nature of the Spiral model enables us to build the game in incremental versions, with each spiral producing a progressively complete and more refined prototype. This allows us to test our onboarding systems with players at multiple points throughout

development, gathering evidence on whether our tutorial and streamlined controls are effectively lowering the barrier to entry. Each iteration brings us closer to the optimal balance between depth and accessibility.

Third, the Spiral model is particularly well-suited for academic research projects with complex innovation goals. A 2025 thesis from the University of Baguio titled "Alas Tres: A Role-playing Game of the Philippine Folklore" successfully followed the Spiral methodology for their 2D game development, noting that it allowed for "incrementing the game's development until the final version is released." This local academic precedent validates our methodological choice and demonstrates its applicability in contexts very similar to ours.

This iterative framework directly supports our Developmental Research Design. Each spiral cycle creates specific deliverables that trace the evolution of our solution. Starting with the foundational requirements analysis, each subsequent cycle builds upon the last, adding features, refining mechanics, and incorporating lessons learned from testing and risk assessment. The final cycles provide the crucial evaluation data for our research, turning the development of Bastion of Genesis into a well-documented process that results in both a playable game and a substantive case study.

1. Determining Objectives – At the start of each cycle, the team identifies the specific goals, features, and performance criteria for this iteration. For the first spiral, this includes defining core mechanics, system requirements, and accessibility targets. Later spirals focus on adding specific features like the Pollution-Hostility System, enemy faction behaviors, or multiplayer refinement. Each cycle begins with a clear understanding of what success looks like.

2. Identifying and Resolving Risks – Before any significant development begins, the team conducts a thorough risk assessment. Potential risks might include technical challenges in implementing LAN multiplayer, balance issues in the Strategic Triad, usability problems in the tutorial system, or performance concerns with the pollution mechanics. Mitigation strategies are developed and, where necessary, prototyped and evaluated. This phase may involve creating small proof-of-concept demonstrations to validate that risky approaches will work before committing to full-scale development.

3. Development and Testing – With risks addressed and mitigated, the team proceeds with implementation, coding the planned features, creating assets, and integrating systems. This phase culminates in comprehensive testing to verify functionality, performance, and alignment with the cycle's objectives. Testing includes both technical validation (bug detection, performance measurement) and user experience evaluation (playability, intuitiveness, engagement).

4. Planning the Next Iteration – Based on test results, risk analysis, and stakeholder feedback, the team evaluates progress and plans the subsequent spiral. This includes defining objectives for the next cycle, incorporating lessons learned, refining the project roadmap, and updating risk assessments. This phase ensures that each spiral builds meaningfully upon the end, with continuous improvement driving the project forward.

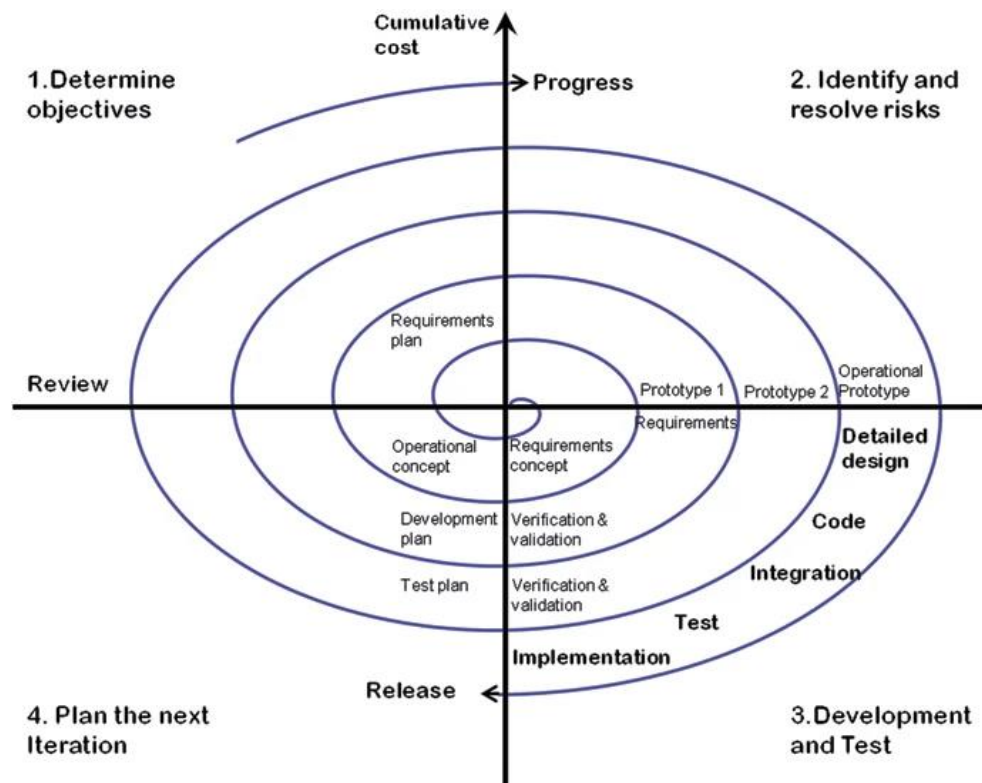


Figure 22. Spiral Model

- **Work breakdown structure**

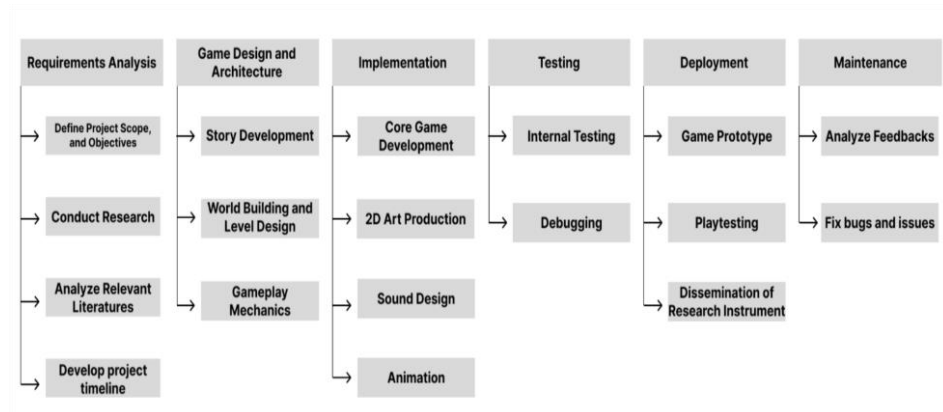


Figure 23. Work Breakdown Structure

The researchers followed a structured Work Breakdown Structure (WBS) based on the Waterfall Method. This method divides the project into sequential phases, ensuring that each step is completed before moving on to the next. Starting with Requirements Analysis, the team identified technical needs, gameplay mechanics, and elements that are necessary for the game. Following this, Game Design and Architecture focused on building the blueprint for core game elements, including level design, and user interface. Next, during the Implementation phase, researchers developed the gameplay mechanics and assets, utilizing tools such as Unity to bring the game world and its interactive elements to life. Testing was then conducted to identify and resolve bugs and ensure functionality met player expectations. The Deployment phase introduced the prototype to a group of play testers, providing valuable feedback for further refinement. Finally, Maintenance involved ongoing updates and adjustments based on user feedback, ensuring a polished experience for future stages of development.

3.2.3 Evaluation Procedures

The evaluation of Bastion of Genesis will be conducted through a structured procedure involving two respondent groups. 10 Game Design and Development Professionals will provide a technical assessment, while 50 target Users (PC Gamers aged 13 and above) will participate in playtesting to evaluate the player experience. Both groups will provide feedback via a tailored survey questionnaire using Likert scales and open-ended questions. Their responses will be measured against five key criteria: Functional Suitability to test core mechanics, Performance Efficiency for technical stability, Usability for player intuitiveness and satisfaction, Compatibility for system operation, and Reliability for game maturity and error handling. This method ensures a comprehensive analysis that validates both the technical execution and player-centric design goals of the project.

- **Data Gathering**

To gather meaningful feedback on Bastion of Genesis, this study will use the Survey Questionnaire Method. This approach allows us to collect organized and consistent insights from all our participants.

The researchers will design a questionnaire with specific questions, using scales to measure how easy and enjoyable the game is to play, alongside open-ended questions to understand each player's unique experience. This method allows for the efficient collection of information that can be analyzed to determine whether the game successfully makes strategy gaming more accessible.

The researchers will select participants using Purposive Sampling. This means they will intentionally invite people who possess the specific knowledge or background that the research requires to provide the most useful feedback.

- For the 10 Game Design and Development Professionals, the researchers will seek out individuals with professional or advanced academic experience in creating games.

Their specialized skills in design and programming are essential for giving us deep, technical advice on how well our game systems work.

- For the 50 Users, the researchers will focus on PC Gamers aged 13 and above. These players represent the exact audience Bastion of Genesis is being made for. Their personal experience with playing games on a computer makes their opinions about the tutorial, controls, and overall fun incredibly valuable for the study.

By using surveys with this focused approach to choosing participants, the research ensures that the feedback received is both relevant and trustworthy, directly supporting the evaluation and improvement of the game.

- **Respondents**

The respondents for this study will consist of two distinct groups, totaling 60 participants, to ensure comprehensive and multi-perspective feedback on Bastion of Genesis.

Table 3. Participants

Participants	No. of Participants
Game Design and Development Professionals	10
PC Gamers aged 13 and up	50

- **Statistical Treatment**

The responses will be measured using the weighted mean. Results from Likert scale type questions.

The formula for weighted mean:

$$mean = \frac{\sum f(x)}{N}$$

where:

f = frequency

x = assigned weighted mean

$\sum f(x)$ = sum of all frequency multiplied by assigned weight

N = total number of respondents

To verbally interpret the weighted mean, the researchers used the 5-point Likert Scale.

Table 4. 5-point Likert Scale

Weight	Verbal Interpretation	Scale
5	Excellent	4.21 - 5.00
4	Very Satisfactory	3.41 - 4.20
3	Satisfactory	2.61 - 3.40
2	Fair	1.81 - 2.60

1	Poor	1.0 - 1.80
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- **User Acceptance Evaluation (ISO/IEC 25010)**

The evaluation of Bastion of Genesis will be guided by key quality characteristics adapted from the ISO/IEC 25010 standard, ensuring a comprehensive assessment of both its technical execution and player experience. The following five criteria have been selected as the most applicable and relevant for a game of this scope:

Table 5. User Acceptance Evaluation (ISO/IEC 25010)

Characteristics
Functional Suitability
Performance Efficiency
Usability
Compatibility
Reliability

The researchers followed a structured Work Breakdown Structure (WBS) based on the **Spiral Model**. This iterative method utilizes repeated cycles to manage project complexity, ensuring that risks are mitigated before development progresses. Starting with **Determining Objectives**, the team identified core survival-strategy mechanics, such as the Strategic Triad, alongside technical requirements. Following this, **Identifying and Resolving Risks** focused on assessing potential imbalances in the Pollution-Hostility System and technical constraints to prevent development setbacks early on. Next, during **Development and Tests**, the researchers utilized Unity to implement game assets and interactive systems, followed by rigorous testing with participants using a weighted mean to evaluate user acceptance. Finally, **Planning the Next Iteration** involved evaluating results and participant feedback to refine the game prototype and update the development roadmap for the subsequent spiral cycle. This approach ensures a polished experience that balances strategic depth with accessible onboarding.